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Allostatic load associated with all-cause and cause-specific mortality: a large prospective cohort study --Manuscript Draft--

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Abstract:	Background Allostatic load (AL) is a composite measure of cumulative physiological burden from chronic stress, quantifying multisystem dysregulation. Whether AL may alter the risk of overall, cause-specific mortality, and premature mortality remains unclear. Methods A total of 378,785 participants (mean age 56.6 years; 53.8% female) were included from the UK Biobank. AL was assessed by integrating 12 commonly applied metrics covering cardiovascular, metabolic, immune, and kidney systems (score range: 0-12), with higher scores indicating greater physiological dysregulation. We examined the associations of AL with all-cause and cause-specific mortality, and assessed the effect modification by age at participants recruitment and at death. Findings In this prospective study, 39,804 deaths were recorded over a median follow-up of 15.4 years, including 14,852 from cancer, 7118 from cardiovascular disease, 1938 from respiratory disease, and 1521 from digestive disease. Higher AL scores were significantly associated with increased risks of all-cause and cause-specific mortality in a dose-dependent manner, with the hazard ratio (95% confidence interval) of 1.11 (1.11-1.12) for all-cause mortality, and 1.08 (1.07-1.09) for cancer, 1.20 (1.18-1.21) for cardiovascular, 1.11 (1.09-1.14) for respiratory, and 1.25 (1.21-1.29) for digestive disease mortality (all P-trend<0.001), per each point increase in AL scores. Age significantly modified these associations (all P-interaction<0.01), with stronger associations for premature mortality (age-at-death<70 years) or individuals with younger age at enrollment. Interpretation These findings establish AL as a promising signature for risk stratification of mortality, underscoring the necessity of implementing preventive interventions among younger adults with heightened chronic physiological stress. Funding See the Acknowledgement.

Allostatic load associated with all-cause and cause-specific mortality: a large prospective cohort study

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Abstract

Background Allostatic load (AL) is a composite measure of cumulative physiological burden from chronic stress, quantifying multisystem dysregulation. Whether AL may alter the risk of overall, cause-specific mortality, and premature mortality remains unclear.

Methods A total of 378,785 participants (mean age 56.6 years; 53.8% female) were included from the UK Biobank. AL was assessed by integrating 12 commonly applied metrics covering cardiovascular, metabolic, immune, and kidney systems (score range: 0-12), with higher scores indicating greater physiological dysregulation. We examined the associations of AL with all-cause and cause-specific mortality, and assessed the effect modification by age at participants recruitment and at death.

Findings In this prospective study, 39,804 deaths were recorded over a median follow-up of 15.4 years, including 14,852 from cancer, 7118 from cardiovascular disease, 1938 from respiratory disease, and 1521 from digestive disease. Higher AL scores were significantly associated with increased risks of all-cause and cause-specific mortality in a dose-dependent manner, with the hazard ratio (95% confidence interval) of 1.11 (1.11-1.12) for all-cause mortality, and 1.08 (1.07-1.09) for cancer, 1.20 (1.18-1.21) for cardiovascular, 1.11 (1.09-1.14) for respiratory, and 1.25 (1.21-1.29) for digestive disease mortality (all P -trend<0.001), per each point increase in AL scores. Age significantly modified these associations (all P -interaction<0.01), with stronger associations for premature mortality (age-at-death<70 years) or individuals with younger age at enrollment.

Interpretation These findings establish AL as a promising signature for risk stratification of mortality, underscoring the necessity of implementing preventive interventions among younger adults with heightened chronic physiological stress.

Funding See the Acknowledgement.

Research in context

Evidence before this study

Clarifying the risk factors associated with death, especially premature death, is of paramount public health importance. Allostatic load (AL), an index of cumulative multisystem dysregulation, has been linked with elevated mortality risk, but evidence concerning its association with overall and cause-specific mortality as well as premature mortality remains limited.

Added value of this study

In this prospective study of 378,785 UK Biobank participants, higher AL scores were significantly associated with increased risks of all-cause mortality and mortality from cancer, cardiovascular, respiratory, and digestive disease in a dose-response manner. These associations appeared stronger for premature mortality at age <70 years.

Implications of all the available evidence

AL may serve as a robust biomarker for mortality, which bears potential public health significance for mortality risk stratification and tailored preventive interventions targeting younger adults with high chronic stress exposure.

Introduction

The global burden of mortality has become increasingly dominated by non-communicable diseases (NCDs), particularly cardiovascular disease (CVD), cancer, respiratory disease, and digestive disease^{1,2}. In 2021, NCDs accounted for at least 43 million deaths globally, including over 18 million premature deaths before age 70^{1,2}. Although premature mortality from cancer is declining in high-income countries³, several NCDs—such as colorectal cancer^{4,5}, type 2 diabetes^{6,7}, and heart failure⁸—are showing trends toward earlier onset and younger-age mortality. These shifting mortality patterns raise concerns about the long-term public health impact and underscore the need for scalable, integrative efforts to identify high-risk individuals.

Chronic exposures, such as unhealthy behaviors, sustained psychosocial stress and socioeconomic disadvantage, can lead to prolonged activation of body's adaptive regulatory systems, a process known as allostasis⁹⁻¹¹. The cumulative physiological cost of maintaining allostasis under chronic stressors is conceptualized as allostatic load (AL), quantified through a composite index of biomarkers across neuroendocrine, metabolic, immune, kidney, and cardiovascular systems^{10,12}. As an integrative measure of multisystem dysregulation, elevated AL could plausibly indicate reduced resilience and impaired adaptive capacity, thereby contributing to increased risk of mortality¹³. Although several prospective studies have examined associations between AL measures and mortality, most have primarily investigated all-cause mortality without adequately addressing specific causes of death, largely due to limited sample sizes and insufficient cause-specific deaths¹⁴⁻¹⁶. Additionally, a few

studies have specifically assessed the association between AL and mortality among specific clinical populations, such as patients diagnosed with breast or colorectal cancer^{17,18}, but the generalizability of the findings to the broader healthy population remains uncertain. Moreover, existing studies have rarely performed age-stratified analyses. Consequently, our knowledge regarding the extra risk of premature deaths associated with increased AL exposure was restricted, and it remains unclear whether the associations between AL and mortality differ by age groups.

Leveraging the large-scale UK Biobank (UKB) cohort with long-term follow-up, this study aimed to examine the associations of AL levels, constructed from 12 biologically relevant and commonly applied metrics, with both all-cause and cause-specific mortality from major disease categories. In addition to estimating the overall mortality, we placed emphasis on the risk of premature deaths and mortality risk among different age groups.

Methods

Study population

The UKB (<https://www.ukbiobank.ac.uk/>, Application No. 90999) is a population-based prospective cohort study comprising over 500,000 individuals aged 40-69 years, which were recruited between 2006 and 2010^{19,20}. Details of the study design and population have been described previously^{19,20}. The UKB received ethical approval from North-West Multi-center research Ethics Committee (reference 11/NW/0382) and all participants provided written informed consent.

We excluded individuals with missing data on any of the 12 metrics used to construct the AL index (n=122,533), and those who died within the first year of follow-up (n=670), leaving a total of 378,785 eligible participants (**appendix p-23**). Specifically, participants with relevant disease at baseline were additionally excluded from the corresponding cause-specific mortality analyses, including 13,141 participants with myocardial infarction, angina or stroke for CVD mortality (final n=365,644); 27,956 with any cancer for cancer mortality (final n=350,829); 2287 with chronic obstructive pulmonary disease (COPD) or emphysema for respiratory disease mortality (final n=376,498); and 623 with liver failure, cirrhosis or alcoholic liver disease for digestive disease mortality (final n=378,162).

Assessment on allostatic load

AL was assessed by combining 12 routinely measured metrics selected for their biological relevance and frequent use in prior AL research^{13,17,18,21}, covering allostatic regulation of the cardiovascular (pulse rate, systolic blood pressure, and diastolic blood pressure), metabolic (glycated hemoglobin, albumin, high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol, triglycerides, and waist-to-hip ratio), immune (C-reactive protein and insulin-like growth factor-1(IGF-1)), and kidney systems (serum creatinine). Each metric was categorized into quartiles based on its distribution in the overall participants, and a score of 1 was assigned for values in the highest quartile (≥ 75 th percentile), except for HDL-C, albumin, and IGF-1, for which a score of 1 was assigned for the lowest quartile (≤ 25 th percentile); all other

values were scored as 0. The definition of the high-risk quartile for each individual biomarker is shown in **appendix p-3**. Scores for 12 metrics were summed to generate a total AL score ranging from 0 to 12, with higher scores reflecting greater multisystem physiological dysregulation.

Assessment of outcomes

During the follow-up, data on mortality were obtained through linkage with national death registries, from NHS England (for England and Wales) and from the NHS Central Register, part of the National Records of Scotland (for Scotland)¹⁹. The International Classification of Diseases, 10th Revision (ICD-10) codes were used to identify the causes of death.

The primary outcomes of our study were all-cause mortality and cause-specific mortality. For cause-specific analyses, we focused on four leading causes to NCD mortality^{1,2}: CVD (I00-I99), cancer (C00-D48), respiratory disease (J30-J98, with exception of J69 and J96), and digestive disease (K00-K93). Additionally, mortality from major individual diseases were evaluated as secondary outcomes based on narrower ICD-10 groupings.

Statistical Analyses

The person-years of follow-up was calculated from the date of recruitment (between 2006 and 2010) until the date of death, loss to follow-up, or September 1st, 2024, whichever occurred first.

We examined the associations of the AL scores, primarily classified as eight categories (score 0, 1, 2, 3, 4, 5, 6, and ≥ 7), with all-cause and cause-specific mortality due to cancer, CVD, respiratory disease, or digestive disease. The highest category (≥ 7) combined scores that individually represented $<5\%$ of the study population. For each disease categories, we considered specific disease cause with at least 50 events to ensure statistical robustness. For all-cause mortality, Cox proportional hazards models were used to estimate the age- and sex-adjusted, and multivariate-adjusted hazard ratios (HRs) and 95% confidence intervals (CIs), adjusting for age, sex, smoking status, alcohol consumption, Townsend deprivation index, household income, educational attainment, ethnicity, medication use, status of healthy diet, and physical activity (**appendix pp 1-2**). For cancer-specific mortality, family history of cancer in first-degree relatives was additionally adjusted for. For cause-specific mortality, Fine-Gray subdistribution hazard models were implemented to account for competing risks from other causes of death. Proportional hazards assumption was evaluated using Schoenfeld residual tests for Cox models and by examining time-by-covariate interactions in Fine-Gray models. When evidence of non-proportionality was detected, stratified analyses were conducted by follow-up time intervals. Dose-response relationships were assessed using the continuous measure of AL scores, for which we calculated the P values for trend (P -trend). Restricted cubic splines were employed to examine the potential non-linear associations.

Subgroup analyses were performed to examine the associations of AL scores

with mortality risk by age at deaths in two (premature mortality, defined as death occurring before 70 years^{2,3,22} or non-premature mortality) or five categories (<50, 50-59, 60-69, 70-79, or ≥ 80 years). Stratified analyses were also conducted by age at participant enrollment or other stratum defined by major covariates. Likelihood ratio tests were used to calculate the *P* values for interactions (*P*-interaction) between AL scores and the subgroup variables.

Several sensitivity analyses were conducted to assess the robustness of findings, including the exclusion of deaths within the first two years of follow-up to mitigate reverse causality, the exclusion of deaths occurring post-COVID-19 pandemic onset to account for the potential period effects, the exclusion of individuals with regular medication use or chronic conditions at baseline to partly address the concern on residual confounding. Analyses were also conducted by restricting to participants of non-White ancestry.

All analyses were conducted using R 4.4.2. Statistical significance was defined as the two-sided $P < 0.05$.

Role of the funding source

The funders had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Baseline characteristics

This study included 378,785 participants for all-cause mortality analysis (mean age:

56.6 years; 53.8% female, **Table-1**). Those with a higher AL score were more likely to be older and males, and tend to have lower educational attainment, reduced household income, and greater socioeconomic deprivation. Individuals in higher AL groups also demonstrated a higher prevalence of behavioral and clinical risk factors, such as smoking, medication use, poor diet status and insufficient physical activity.

Scores of AL and risk of all-cause mortality and cause-specific mortality from major disease categories

During a median follow-up of 15.4 years (IQR: 14.6-16.1 years), we documented 39,804 all-cause deaths, among which 14,852 cancer deaths, 7118 CVD deaths, 1938 respiratory disease deaths and 1521 digestive disease deaths were identified.

Participants who died during follow-up had a higher AL score compared to those who remained alive (**appendix p-4**). A higher AL score was associated with elevated risks of all-cause mortality. Compared with individuals with score of 0, the multivariate-adjusted HR (95% CI) for all-cause mortality was 1.12 (1.05-1.19) for a score of 1 and 2.17 (2.04-2.31) for scores ≥ 7 , displaying an upward trend in mortality with increasing AL scores (HR=1.11, 95%CI: 1.11-1.12 per one point of score increase, P -trend<0.001, **Table-2**). Restricted cubic spline models revealed no significant deviation from linearity of the associations (**appendix p-24**). AL scores were also significantly associated with cause-specific mortality from four major disease categories in a dose-dependent manner, with a multivariate-adjusted sHR (95%CI) of 1.08 (1.07-1.09) for cancer, 1.20 (1.18-1.21) for CVD, 1.11 (1.09-1.14) for respiratory

disease, and 1.25 (1.21-1.29) for digestive disease mortality per one point of score increase (all P -trend<0.001). Because the proportional hazards assumption was not satisfied for CVD and respiratory disease mortality (P <0.05; **appendix p-5**), follow-up time-stratified Fine-Gray models were further applied, which yielded consistent associations for a follow-up time <10 years or $\geq$ 10 years (**appendix pp 6-7**). These associations remained similar when AL was categorized into quartiles (**appendix p-8**). Analysis of AL components revealed that each of the four components were significantly associated with both all-cause and cause-specific mortality, with the metrics related to inflammatory and kidney function exhibiting the strongest associations for all outcomes (**appendix p-25**).

AL scores and cause-specific mortality from individual diseases

We examined the risk of mortality due to major individual diseases (**Figure-1** and **appendix pp 9-10**). Among cancer deaths, increasing risks were found for all individual cancers of the digestive system, with the highest effect estimates observed for liver cancer (sHR=1.27, 95%CI: 1.21-1.33 per one point of score increase), and bile duct and gallbladder cancer (sHR=1.23, 95%CI: 1.12-1.36). Significantly increased risk of cancer mortality was also demonstrated for kidney cancer (sHR=1.18, 95%CI: 1.11-1.25), as well as for lung cancer, breast cancer, uterine cancer, prostate cancer, bladder cancer and leukemia, even after FDR correction. In addition, an elevated risk was observed for myelodysplastic syndrome (sHR=1.19, 95%CI: 1.04–1.35), whereas no significant associations were noted for other non-

malignant neoplasms. For mortality due to digestive diseases, the highest effect estimates were also concentrated in chronic liver diseases, such as liver fibrosis and cirrhosis (sHR=1.52, 95%CI: 1.37-1.70), and alcoholic liver disease (sHR=1.39, 95%CI: 1.30-1.49). Similarly, higher AL scores were associated with elevated risk of mortality from major CVDs such as chronic ischemic heart disease (sHR=1.24, 95%CI: 1.21-1.27), acute myocardial infarction (sHR=1.24, 95%CI: 1.20-1.28), hypertensive heart disease (sHR=1.26, 95%CI: 1.16-1.36), and ischemic stroke (sHR=1.24, 95%CI: 1.14-1.34). For respiratory diseases, elevated AL scores were significantly associated with deaths from COPD (sHR=1.05, 95%CI: 1.01-1.08), bronchiectasis (sHR=1.14, 95%CI: 1.03-1.26) and interstitial pulmonary diseases (sHR=1.21, 95%CI: 1.16-1.25).

Sensitivity analysis

Sensitivity analyses excluding deaths occurring within the first two years of follow-up, deaths post COVID-19 pandemic, participants with baseline medication use, those with pre-existing chronic diseases, or non-White ethnicity did not materially change the findings either for all-cause mortality, cause-specific mortality due to major disease categories (**appendix pp 11-13**) or from individual diseases (data not shown).

Subgroup analyses by age at death

We further examined AL scores associated with mortality stratified by age at death.

Cumulative risk of mortality curves began to diverge by age 60, and the separation

became pronounced by age 70, with higher scores showing progressively greater risks (**appendix p-26**). By age 85, the cumulative risk of mortality was 25% for score 0 and 49% for score 7, highlighting a marked absolute risk difference. The association with all-cause mortality was consistently stronger for premature deaths (<70 years) than for deaths at ≥ 70 years (P -interaction<0.001; **Table-3 and appendix pp 14-15**); the HR (95%CI) per one point of score was 1.17 (1.16-1.18) and 1.10 (1.09-1.11) respectively. Further stratified analysis by 10-year intervals of age at death corroborated the findings, which revealed a trend towards stronger associations for deaths at a younger age group (**Table-3**).

Similar to the pattern observed in all-cause mortality, a comparable shift in the risk profile was also noted for mortality from all major disease categories, with a more pronounced association estimate observed for deaths occurring at a younger age (**Table-3 and appendix pp 14-15**). The per-point-of-score sHR (95%CI) for premature vs. later deaths was 1.10 (1.09-1.12) vs. 1.06 (1.06-1.07) for cancer, 1.28 (1.25-1.30) vs. 1.16 (1.15-1.18) for CVD, 1.20 (1.14-1.26) vs. 1.09 (1.06-1.12) for respiratory disease, and 1.32 (1.26-1.38) vs. 1.22 (1.17-1.26) for digestive disease (all P -interaction<0.05). Notably, AL scores exhibited stronger associations with premature mortality from several individual diseases, particularly most CVDs such as acute myocardial infarction, chronic ischemic heart disease, and hemorrhage stroke (**appendix pp 16-17**; all P -interaction<0.05).

Subgroup analyses by age at participant enrollment

A clear increase in all-cause mortality risk was observed with elevated AL levels and age at participant enrollment, in a dose-dependent manner. The HR (95%CI) for all-cause mortality was 2.24 (2.01-2.50) for AL score ≥ 5 in those aged <50 years, 4.74 (4.34-5.17) in those aged 50-60 years, and 10.16 (9.36-11.03) in those aged ≥ 60 years. The association between AL scores and all-cause mortality appeared strongest in participants aged <50 years (HR=1.22 per one-unit increase, 95%CI: 1.19-1.24) at cohort enrollment, followed by those aged 50-60 (HR=1.17, 95%CI: 1.15-1.18) and ≥ 60 years (HR=1.11, 95%CI: 1.10-1.12), suggesting an effect modification of age at enrollment on the association between AL scores and all-cause mortality (P -interaction <0.001 ; **Figure-2**). Similar statistical interaction was also found for cancer, CVD, and digestive disease mortality respectively (all P -interaction <0.05 ; **Figure-2**). A similar pattern of particularly stronger associations at younger ages was also observed for mortality from several individual diseases, such as esophageal cancer, gastric cancer, bladder cancer, acute myocardial infarction, chronic ischemic heart disease, and vascular disorders of the intestine (P -interaction <0.05 ; **appendix pp 18-19**).

Subgroup analyses by other demographic and lifestyle factors

Associations between AL scores and all-cause and cause-specific mortality were generally consistent across all subgroups (**appendix pp 20-22**). No substantial effect modification by demographic factors was observed for all-cause mortality, but stronger associations were noted among females for respiratory and digestive disease

mortality (P -interaction <0.05). Among lifestyle factors, associations with all-cause mortality were less pronounced in current smokers, with a similar effect modification by smoking status found for respiratory and digestive disease mortality (P -interaction <0.05). Additionally, a more pronounced effect magnitude was observed for CVD mortality among individuals with higher annual household income, and for respiratory disease mortality among those with higher educational attainment.

Discussion

Leveraging data from a large population-based cohort of 378,785 individuals, we found that higher AL scores were associated with progressively elevated risk of all-cause mortality and mortality from major disease categories, including cancer, cardiovascular, respiratory, and digestive disease. Associations were particularly pronounced for premature mortality (death at age <70 years) and among individuals with a younger age at enrollment, underscoring the potential impact of early-life stress and cumulative physiological burden on long-term survival.

Our study demonstrates the potential of AL scores in capturing interindividual variations in biological vulnerability and resilience. The robust associations between scores of AL and mortality risks in our study consistently reflect the cumulative toll of chronic stress on physiological systems^{13,23}. Perceived stress activates the hypothalamic-pituitary-adrenal axis and related neuroendocrine pathways, resulting in glucocorticoid dysregulation, and structural changes in stress-regulating brain regions^{24,25}. These neuroendocrine changes drive systemic metabolic imbalance and

mitochondrial dysfunction, which promote inflammation, epigenetic reprogramming, and cellular aging, and has been conceptualized as “mitochondrial AL”²⁶.

Subgroup analyses confirmed that AL scores were more strongly associated with premature mortality, consistent with evidence that environmental and socioeconomic exposures explain substantially more variation in early mortality than polygenic risk^{27,28}. In this context, AL can be viewed as a biological indicator of the exposome, capturing cumulative physiological burden across systems, which provides a mechanistic rationale for the comparatively stronger AL-premature mortality associations in our study. Consistent with this pattern, stratified analyses by baseline age demonstrated significant associations between AL scores and mortality risk across all age categories, with relative risks being most pronounced among younger adults. While a prior meta-analysis reported stronger associations between AL and all-cause mortality in studies with a mean age below 65 years¹⁶, but that study lacked individual-level age data and was thus unable to investigate the associations among individuals within specific age subgroups. Leveraging a well-established prospective mega-cohort with long-term follow-up, our current study was able to refine the age gradient and demonstrate particularly pronounced associations in participants younger than 50 years. These findings highlight AL as a signature of early physiological dysregulation and a modifiable predictor of premature mortality, underscoring the importance of early-life interventions targeting elevated AL scores.

Mortality risks of major NCDs^{1,2} were positively associated with elevated AL scores. For cancer mortality, associations were particularly strong for liver, biliary and

kidney cancers, and elevated AL scores were also related to deaths from non-malignant liver diseases. These findings point to heightened susceptibility of hepatobiliary and kidney systems to cumulative physiological burden from chronic stress, which has been supported by experimental evidence implicating ferroptosis²⁹, mitochondrial dysfunction³⁰, and immune-mediated hepatic injury in stress-related liver pathology³¹, as well as mitochondria-centered pathways in kidney disease biology³². At a broader oncologic level, stress perturbs neuroendocrine-immune circuits and reshapes metabolic and microbiota milieus that influence tumor evolution, dissemination, and therapeutic response, providing additional pathways through which elevated AL may relate to cancer mortality³³. Among non-malignant neoplasms, AL scores also showed stronger associations with myelodysplastic syndrome, a disorder characterized by redox imbalance or oxidative stress and immune-inflammatory dysregulation that can drive genomic instability and clonal evolution³⁴.

Elevated AL scores were associated with increased mortality across a wide range of CVDs, with particularly strong associations observed for leading causes of deaths such as chronic ischemic heart disease, acute myocardial infarction, and ischemic stroke. Besides, we observed more pronounced associations of AL components reflecting inflammation and kidney function with CVD mortality than of other components. This pattern is consistent with evidence that stress-related neural activity, inflammaging, and trained immunity may sustain vascular inflammation and plaque vulnerability, thereby exerting cumulative stress to CVD mortality³⁵⁻³⁷. In parallel,

AL-related kidney dysfunction can amplify risk through inflammation- and oxidative stress–driven fibrosis and cardiorenal interactions³⁸. Notably, elevated AL scores were more strongly associated with premature mortality from several individual CVDs, emphasizing their particular susceptibility to stress-related burden in younger populations.

Despite overall declines in NCD mortality in some regions, the burden of premature deaths—particularly those considered avertable—remains substantial and unevenly distributed^{39,40}. A proportion of such deaths can be prevented through effective policies and health care, but the extent of preventability varies by disease categories and settings⁴⁰. In this context, AL has the potential to offer added value to population risk stratification and surveillance, which may be achieved by pinpointing subgroups more prone to experience avertable premature deaths, especially among younger individuals at high risk. Beyond risk prediction, AL may also serve as a traceable surrogate endpoint reflecting cumulative stress-related dysregulation, providing a framework to evaluate the efficacy of interventions targeting stress-related pathways and refining prevention strategies.

Several limitations should be acknowledged. First, AL scores were derived from a single time-point measurement, precluding evaluation of longitudinal changes that may offer more nuanced insights into dynamic physiological dysregulation¹⁵. Second, although sociodemographic and behavioral variables were adjusted for, genetic predisposition to cause-specific mortality was incompletely addressed, given the varying heritability across disease outcomes. Notably, only cancer-specific models

included family history as a proxy. Future studies should explore gene-AL score interactions to enhance understanding of individualized risk. Third, the UKB cohort predominantly comprises individuals of European ancestry, limiting generalizability to other populations due to smaller non-European samples. Finally, cause-specific analyses were confined to mortality from four major NCD categories. Given the comparatively modest number of events, other crucial NCDs, such as diabetes, were not included in the analyses.

In conclusion, elevated AL scores were consistently associated with increased all-cause and cause-specific mortality, with particularly strong effects in younger individuals and for premature deaths. These findings support AL as a modifiable indicator of early physiological dysregulation and highlight its potential utility in identifying high-risk individuals. Incorporating AL scores into risk stratification frameworks may enhance prevention strategies for negative health consequences and premature deaths, especially among younger populations.

Contributors: HM Xu and WQ Li conceived and designed the study. HM Xu drafted the manuscript. HM Xu, ZC Liu, XL Wang, YZ, TZ, and JY Zhang contributed to data collection and analysis. KF Pan provided critical revisions. WQ Li supervised the study and is the guarantor of the study. All authors reviewed and approved the final manuscript.

Data sharing: Data are available in the UK Biobank upon application (www.ukbiobank.ac.uk). No additional data are available.

Declaration of interests: The authors declare no competing interests.

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Figure legends

Figure 1. Subdistribution hazard ratios for cause-specific mortality from major individual diseases by allostatic load scores. Causes of death with at least 50 events were considered for the analyses. Fine-Gray model was used to assess the associations between score of allostatic load and cause-specific mortality, adjusted for age, sex, smoking status (never, former, or current), frequency of alcohol consumption (never or occasionally, <3 times/week, or ≥ 3 times/week), Townsend Deprivation Index, annual household income (<£18,000, £18,000–30,999, £31,000–51,999, or $\geq$ £52,000), education level (basic education or below, vocational or A/AS-level equivalent, or college/university degree), ethnicity (White or non-White), medication use (never or current), status of healthy diet (poor, fair, or ideal), and physical activity level (low, moderate, or high). For cancer-specific mortality, additional adjustment was made for family history of cancer in first-degree relatives. Abbreviations: CI, confidence interval; sHR, subdistribution hazard ratio.

Figure 2. Association between allostatic load scores and all-cause mortality and cause-specific mortality from major disease categories across subgroups by age at recruitment. AL scores were categorized into three groups: 0-1 (Quartile 1), 2-4 (Quartile 2-3), and ≥ 5 (Quartile 4). Participants were stratified by age at baseline (<50, 50-60, and ≥ 60 years). Analyses were conducted using multivariable-adjusted Cox proportional hazards for all-cause mortality and Fine-Gray competing risk models for cause-specific mortality, respectively. The lowest AL group (0-1) in participants aged <50 served as the common reference. We additionally modeled

allostatic load (per score increase), stratified by age groups (<50, 50-60, and ≥ 60 years). *P* for interaction between AL and age group was derived from a multiplicative interaction term in Cox or Fine-Gray model including all covariates and treated allostatic load scores as a continuous variable. Abbreviations: CI, confidence interval; CVD, cardiovascular disease; sHR/HR, (subdistribution) hazard ratio.

Table 1. Baseline characteristics of study participants by scores of allostatic load

Characteristic	Scores of allostatic load							
	0	1	2	3	4	5	6	≥7
N	36,124	64,805	72,381	66,924	54,419	39,315	24,553	20,264
Age, year, mean (SD)	51.7 (7.8)	54.3 (8)	56.4 (8)	57.5 (7.9)	58.2 (7.7)	58.6 (7.5)	58.8 (7.4)	59 (7.3)
Townsend deprivation index, mean (SD)	-1.55 (2.9)	-1.54 (2.9)	-1.48 (3.0)	-1.39 (3.0)	-1.28 (3.1)	-1.11 (3.2)	-0.93 (3.2)	-0.73 (3.3)
Female, %	77.8	68.6	60.4	51.8	44.3	38.3	33.0	27.2
Ethnicity, White, %	88.7	88.6	88.8	88.6	88.4	88.3	88.2	88.4
Education, college or university degree or equivalent, %	44.1	39.5	35.4	31.7	29.2	26.3	24.7	22.2
Annual household income ≥£52,000/year, %	32.9	27.7	23.3	20.7	18.6	16.7	14.9	13.7
Current smoking, %	7.9	8.3	9.4	10.3	11.3	12.5	14	15.8
Alcohol intake of ≥3 times/week, %	46.8	45.9	44.9	43.8	42.6	41.2	39.6	37.9
Current medication use, %	17.4	18.4	19.8	21.5	23.2	24.4	25.6	26.2
Health diet status, poor, %	31.4	33.6	35.9	39.2	42.4	44.9	48.5	52.0
Physical activity level, low %	11.4	12.0	12.7	13.9	15.3	17.0	18.7	21.4
Family history of cancer in first degree relatives, %	32.9	34.5	35.3	35.3	35.2	35.9	35.6	35.3

Table 2. Association of scores of allostatic load with all-cause and cause-specific mortality^a

	No. of participants	No. of deaths	Crude mortality (per 100,000 person-years)	Age- and sex-adjusted sHR/HR (95% CI)	Multivariate-adjusted sHR/HR (95% CI) ^b
All-cause mortality					
Score categories					
0	36,124	1359	246.2	Ref	Ref
1	64,805	3681	373.9	1.16 (1.09-1.23)	1.12 (1.05-1.19)
2	72,381	5824	533.9	1.35 (1.27-1.43)	1.25 (1.17-1.32)
3	66,924	6989	698.7	1.58 (1.49-1.68)	1.40 (1.32-1.48)
4	54,419	6896	855.3	1.79 (1.68-1.90)	1.52 (1.43-1.61)
5	39,315	6112	1060.4	2.13 (2.01-2.26)	1.72 (1.62-1.83)
6	24,553	4466	1254.4	2.45 (2.30-2.61)	1.90 (1.78-2.02)
≥7	20,264	4477	1548.6	2.97 (2.79-3.16)	2.17 (2.04-2.31)
Per one point of score increase	378,785	39,804	703.8	1.17 (1.16-1.17)	1.11 (1.11-1.12)
<i>P</i> -trend				<0.001	<0.001
Cancer mortality^c					
Score categories					
0	34,244	658	125.5	Ref	Ref
1	60,743	1668	180.4	1.13 (1.04-1.24)	1.11 (1.01-1.22)
2	67,173	2360	232.3	1.22 (1.12-1.33)	1.16 (1.06-1.26)
3	61,629	2617	282.8	1.34 (1.23-1.46)	1.24 (1.14-1.35)
4	49,910	2492	335.2	1.47 (1.35-1.61)	1.32 (1.21-1.45)
5	36,189	2180	408.5	1.72 (1.57-1.88)	1.50 (1.37-1.64)
6	22,401	1477	451.6	1.84 (1.67-2.02)	1.55 (1.41-1.71)
≥7	18,540	1400	525.0	2.06 (1.87-2.27)	1.67 (1.52-1.84)
Per one point of score increase	350,829	14,852	282.3	1.11 (1.10-1.12)	1.08 (1.07-1.09)
<i>P</i> -trend				<0.001	<0.001
Cardiovascular disease mortality					
Score categories					
0	35,696	144	26.4	Ref	Ref
1	63,550	479	49.6	1.41 (1.17-1.70)	1.36 (1.13-1.64)
2	70,466	890	83.7	1.90 (1.59-2.26)	1.75 (1.46-2.08)
3	64,503	1192	123.4	2.44 (2.05-2.91)	2.15 (1.81-2.56)
4	51,958	1264	163.7	2.93 (2.46-3.49)	2.49 (2.09-2.96)
5	37,254	1187	216.4	3.62 (3.04-4.32)	2.94 (2.46-3.51)
6	23,120	906	268.8	4.27 (3.57-5.11)	3.30 (2.76-3.96)
≥7	19,097	1056	384.6	5.81 (4.86-6.94)	4.26 (3.56-5.10)
Per one point of score increase	365,644	7118	117.0	1.25 (1.24-1.27)	1.20 (1.18-1.21)

<i>P</i> -trend				<0.001	<0.001
Respiratory disease mortality					
Score categories					
0	36,089	41	7.4	Ref	Ref
1	64,675	120	12.2	1.18 (0.83-1.69)	1.10 (0.77-1.57)
2	72,141	244	22.4	1.69 (1.21-2.36)	1.41 (1.01-1.97)
3	66,546	350	35.2	2.30 (1.65-3.19)	1.76 (1.27-2.45)
4	53,997	357	44.6	2.62 (1.89-3.65)	1.85 (1.33-2.58)
5	38,923	333	58.3	3.20 (2.30-4.46)	2.07 (1.48-2.88)
6	24,220	244	69.3	3.63 (2.59-5.10)	2.13 (1.52-3.00)
≥7	19,907	249	87.4	4.34 (3.09-6.09)	2.29 (1.63-3.23)
Per one point of score increase	376,498	1938	34.4	1.22 (1.19-1.25)	1.11 (1.09-1.14)
<i>P</i> -trend				<0.001	<0.001
Digestive disease mortality					
Score categories					
0	36,116	28	5.1	Ref	Ref
1	64,767	101	10.3	1.63 (1.07-2.48)	1.54 (1.01-2.35)
2	72,298	142	13.0	1.75 (1.16-2.63)	1.54 (1.03-2.32)
3	66,814	256	25.6	3.11 (2.09-4.62)	2.59 (1.75-3.85)
4	54,290	262	32.6	3.67 (2.46-5.46)	2.90 (1.95-4.32)
5	39,209	267	46.4	4.98 (3.34-7.42)	3.68 (2.47-5.49)
6	24,482	230	64.7	6.70 (4.48-10.02)	4.66 (3.11-6.98)
≥7	20,186	235	81.5	8.08 (5.40-12.11)	5.22 (3.47-7.84)
Per one point of score increase	378,162	1521	26.9	1.33 (1.29-1.37)	1.25 (1.21-1.29)
<i>P</i> -trend				<0.001	<0.001

^a Cox proportional hazards models were used for all-cause mortality, while Fine-Gray subdistribution hazard models were used for cause-specific mortality to account for competing risks.

^b Adjusted for age, sex, smoking status (never, former, or current), frequency of alcohol consumption (never or occasionally, <3 times/week, or ≥3 times/week), Townsend Deprivation Index, annual household income (<£18,000, £18,000–30,999, £31,000–51,999, or ≥£52,000), education level (basic education or below, vocational or A/AS-level equivalent, or college, university degree or equivalent), ethnicity (White or non-White), medication use (never or current), status of healthy diet (poor, fair, or ideal), and physical activity level (low, moderate, or high).

^c Additionally adjusted for family history of cancer among first-degree relatives. Abbreviations: AL, allostatic load; CI, confidence interval; HR, hazard ratio; sHR, subdistribution hazard ratio.

Table 3. Associations between allostatic load score and mortality risk stratified by age at death

Age at death ^a	No. of participants	No. of deaths	Crude mortality (per 100,000 person-years)	sHR/HR (95% CI) ^b	P-int
All-cause mortality					
Premature mortality	352,907	13,926	259.7	1.17 (1.16–1.18)	<0.001
<50	339,365	384	7.3	1.19 (1.12–1.25)	
50-59	344,205	2980	55.9	1.20 (1.18–1.22)	
60-69	341,961	10,562	200.1	1.17 (1.15–1.18)	
Non-premature mortality	364,859	25,878	466.3	1.10 (1.09-1.11)	
70-79	349,543	20,654	386.8	1.11 (1.10–1.12)	
≥80	359,635	5224	95.4	1.08 (1.07–1.10)	
Cancer mortality					
Premature mortality	342,036	6059	117.3	1.10 (1.09-1.12)	<0.001
<50	336,096	119	2.3	1.12 (1.01-1.24)	
50-59	337,288	1311	25.6	1.14 (1.10-1.17)	
60-69	340,606	4629	89.8	1.10 (1.08-1.12)	
Non-premature mortality	344,770	8793	168.6	1.06 (1.06-1.07)	
70-79	343,511	7534	145.0	1.07 (1.05-1.08)	
≥80	337,236	1259	24.5	1.06 (1.02-1.09)	
Cardiovascular disease mortality					
Premature mortality	360,916	2390	44.1	1.28 (1.25-1.30)	<0.001
<50	358,574	48	0.9	1.38 (1.18-1.62)	
50-59	359,020	494	9.1	1.32 (1.26-1.39)	
60-69	360,374	1848	34.1	1.27 (1.23-1.30)	
Non-premature mortality	363,254	4728	86.7	1.16 (1.15-1.18)	
70-79	362,204	3678	67.6	1.17 (1.15-1.20)	
≥80	359,576	1050	19.4	1.14 (1.10-1.18)	
Respiratory disease mortality					
Premature mortality	375,040	480	8.6	1.20 (1.14-1.26)	0.0048
<50	374,566	6	0.1	1.28 (0.70-2.32)	
50-59	374,623	63	1.1	1.24 (1.08-1.43)	
60-69	374,971	411	7.3	1.19 (1.13-1.26)	
Non-premature mortality	376,018	1458	25.9	1.09 (1.06-1.12)	
70-79	375,686	1126	20.0	1.08 (1.05-1.12)	
≥80	374,892	332	5.9	1.12 (1.05-1.20)	
Digestive disease mortality					
Premature mortality	377,191	550	9.8	1.32 (1.26-1.38)	<0.001
<50	373,736	16	0.3	1.33 (0.99-1.80)	
50-59	373,823	103	1.8	1.32 (1.18-1.48)	
60-69	374,311	591	10.5	1.19 (1.14-1.25)	
Non-premature mortality	377,612	971	17.2	1.22 (1.17-1.26)	
70-79	375,289	1569	27.9	1.09 (1.06-1.12)	

≥ 80	374,219	499	8.9	1.11 (1.05-1.17)
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^aPremature death was defined as death before age 70. Death age-specific stratification was performed using 10-year intervals.

^bHazard ratios (HRs) and subdistribution hazard ratios (sHRs) were estimated using Cox or Fine-Gray regression models, adjusted for age, sex, smoking status (never, former, or current), frequency of alcohol consumption (never or occasionally, <3 times/week, or ≥ 3 times/week), Townsend Deprivation Index, annual household income (<£18,000, £18,000–30,999, £31,000–51,999, or $\geq$ £52,000), education level (basic education or below, vocational or A/AS-level equivalent, or college, university degree or equivalent), ethnicity (White or non-White), medication use (never or current), status of healthy diet (poor, fair, or ideal), and physical activity level (low, moderate, or high). For cancer-specific mortality, additional adjustment was made for family history of cancer.

Figure 1. Subdistribution hazard ratios for cause-specific mortality from major individual diseases by allostatic load scores

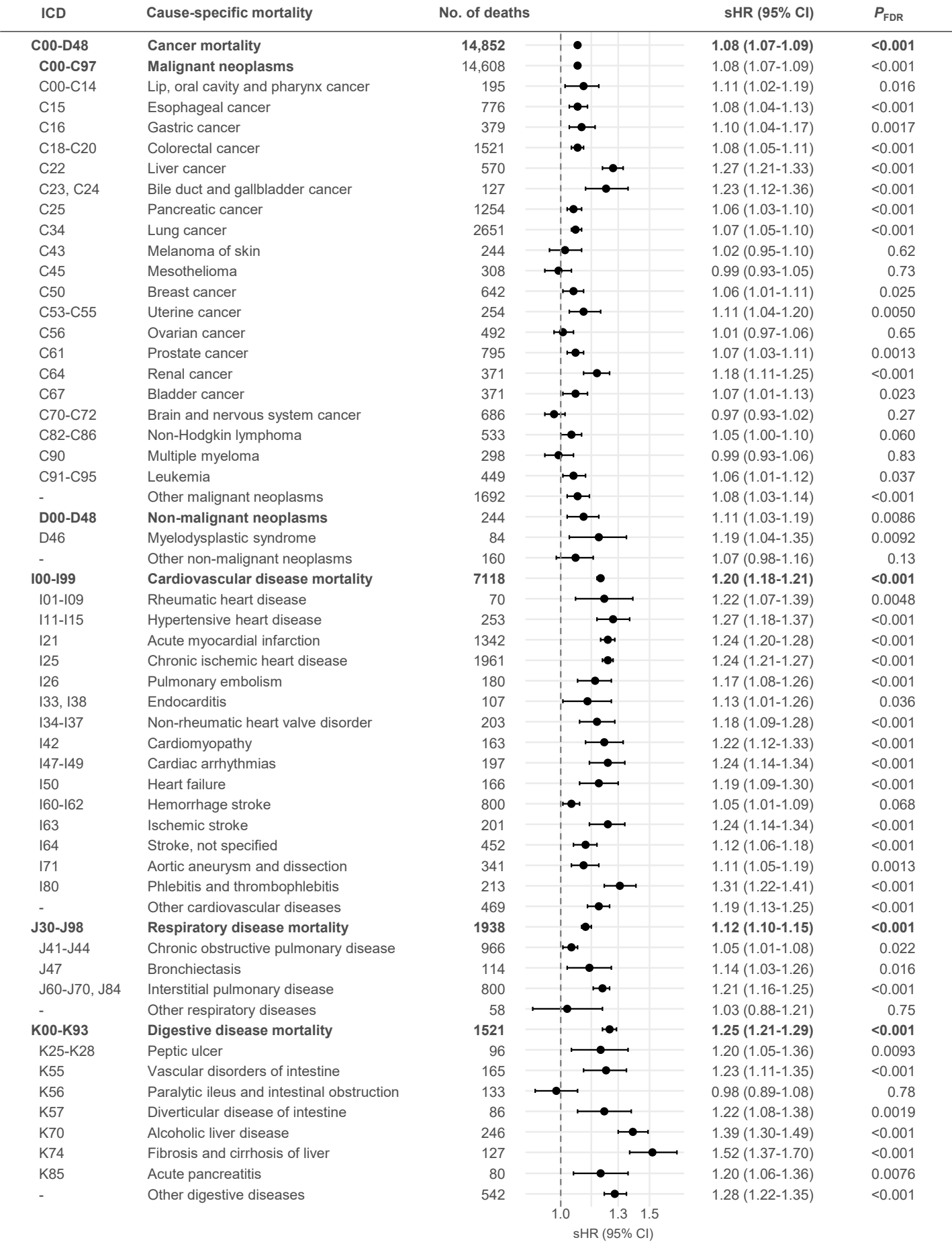
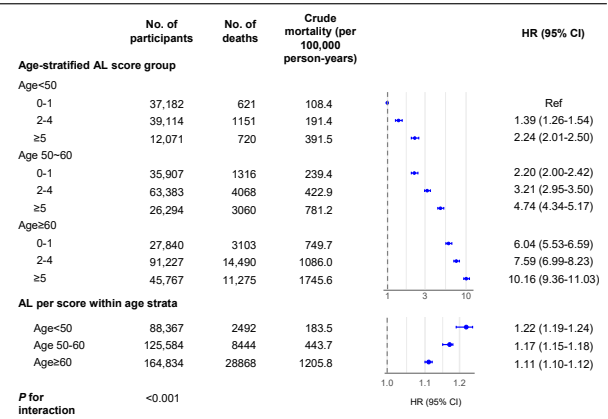
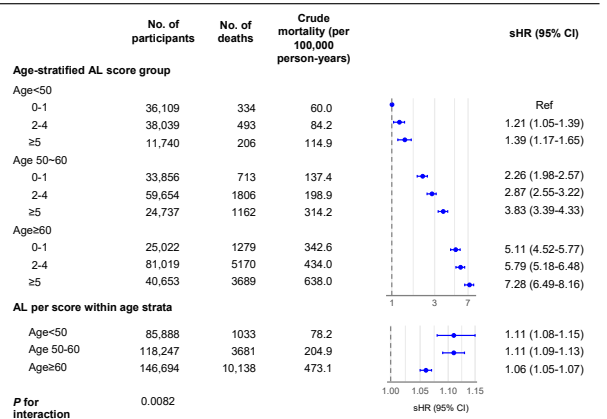


Figure 2. Association between allostatic load scores and all-cause mortality and cause-specific mortality from major disease categories across subgroups by age at recruitment

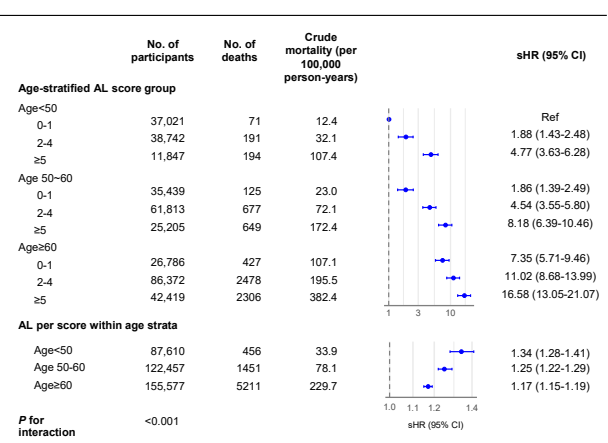
(a) All-cause mortality



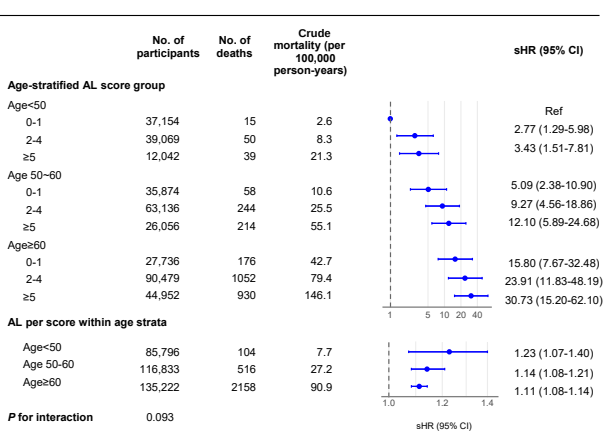
(b) Cancer mortality



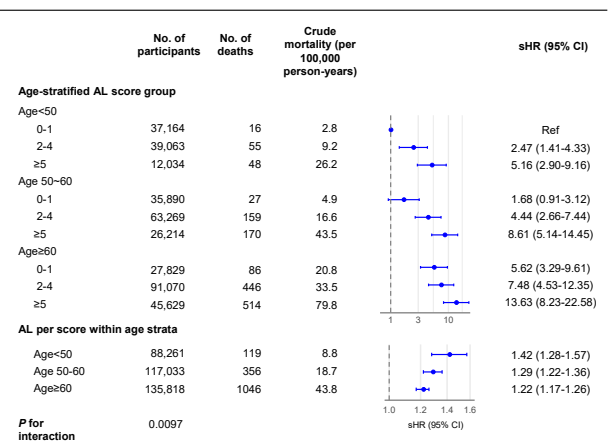
(c) Cardiovascular disease mortality



(d) Respiratory disease mortality



(e) Digestive disease mortality





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